Exercise on Equation Solutions with maximum constraint

- Suppose that a cookie shop has 3 different kinds of cookies. However, some cookie box (say kind 3) contains only 2 cookies. In how many different ways can 5 cookies be chosen, so that you have at most 2 cookies of kind 1?
- **Solution**
  From previous slides of "Exercises on Equation Solutions", without any constraint, the number of solutions is $C(3+5-1,5) = C(7,5) = 21$.
- The number of solutions for constraint $x_3 \geq 3$ is
- $C(3+2-1,2) = C(4,2) = \frac{4!}{2!2!} = 6$
- Thus, the number of solutions when $x_3 < 3$ is $C(7,5) - C(4,2) = 21-6 = 15$.